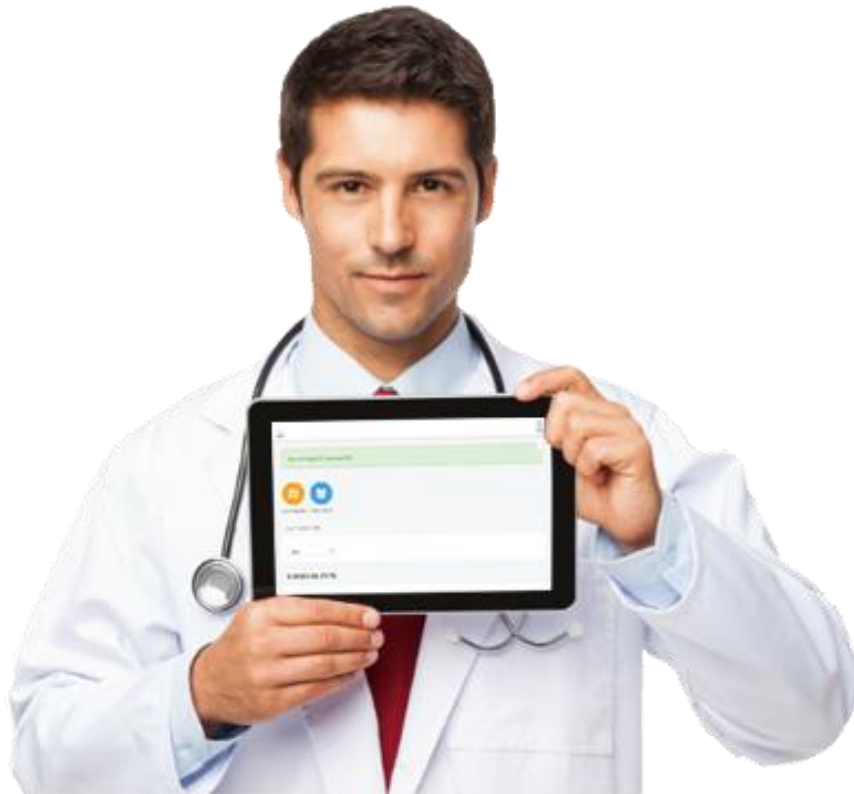




## *Pohnpei Community Health Center of Micronesia goes Paper Less*



Deployment - Customization - Training – Support  
by ViSolve Inc

## Customer Background

PCHC (Pohnpei Community Health Center), a federally funded clinic, was challenged with the task of going digital with limited internet connectivity. PCHC needed a customized system that is scalable, easy to use and that can enable better operational efficiency in terms of handling the medical documents and improve overall productivity at their center. With the help of PIHOA (Pacific Island Health Officers Association) and with a passionate support to Open Source, ViSolve customized OpenEMR for PCHC's deployment.

PCHC transitioned successfully into a paperless environment, enabling digitized workflow all the way from front office registration, dispensing medicines, processing the bills to checking out the patient from the clinic. The clinic offers services relate to general illness, dental & eye programs, cancer, family planning, immunization programs, maternal and child related health programs with regulated pharmacy and laboratory within the clinic.

### Pohnpei Community Health Center



## **Business Problem**

PCHC (Pohnpei Community Health Center) was looking for a customized system that is scalable, easy to use and that can enable better operational efficiency in terms of handling the medical documents and improve overall productivity. The key was to replace paper records to electronic format and digitize the entire clinic. Processing the patient medical documents and searching for medical records from a bunch of paper records during an emergency was cumbersome and time-consuming. By making the entire workflow, all the way from patient registration to patient checkout, electronic, it was assumed that it would enable better outcomes for the clinic. Another major challenge was to educate and train the doctors with limited computer knowledge to become proficient EMR (Electronic Medical Record) users.

## **Proposed Solution**

Based on the business problem, PCHC in assistance with PIHOA (Pacific Island Health Officers Association) came up with a proposal in making use of OpenEMR for handling the patient medical records electronically. ViSolve's long term Open Source Contributions with OpenEMR and Healthcare domain expertise with training and support, helped understand PCHC's business challenges and to provide a customized solution to best fit their needs. ViSolve took up the challenge in helping the clinic to transition from paper records to electronic format. This included onsite Engineering assistance at PCHC to understand customer needs better and accomplish the proposed solution.

The OpenEMR instance was installed within the Clinic's server and all basic configurations that is required for a full fledged clinic to make use of OpenEMR covering the Clinic's regular activities. After a successful configuration, a groups for Doctors and Clinicians were formed. Each group was trained individually on OpenEMR by ViSolve as per the roles and Access privileges were provided within the clinic. Continuous follow-up and monitoring for several weeks helped address post deployment issues faced by OpenEMR users at the Clinic.

Finally, PCHC transitioned successfully into a paperless enabling digitized workflow all the way from front office registration, dispensing medicines, processing the bills to checking out the patient from the clinic.

Customizations related to PCHC forms, UDS reports as per PCHC, and other forms were customized and integrated as a part of OpenEMR such that all the forms can be accessible from the EMR and all paper forms were reduced.

## OpenEMR Installation and Deployment:

As an initial part, a dedicated team from ViSolve with a Remote team and an onsite Engineer was assigned to handle the OpenEMR installation. The deployment and installation was completed with a day of time and all related issues and dependencies related to installation was taken care by ViSolve's Onsite engineer in smoother process making the application available for doctors and admin users within the clinic.

### Customized OpenEMR, Installed for Pohnpei Community Health Center



## ViSolve Service Performance Highlights in helping PCHC going paperless:

**UDS Report:** PCHC has its own set of official reports that was used based on the patient encountered within the clinic. These reports were generated manually during end of financial year that involved lot of manual work in fetching the records and submitting the data. To eliminate the cumbersome process with manual report generation, a new set of reports as custom reports was created under OpenEMR reports and all data specified as per the provided requirement was developed and added with EMR. This reduced the complexity involved in creating the reports manually and the administrative users were able to generate reports with few clicks making it simpler

## Customized OpenEMR (Open Source based) for Pohnpei Community Health Center

| Clinical Indicators                                                                                                                                                     |                                  |                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------|
| Target Date: 2016-12-01                                                                |                                  |                            |
|   |                                  |                            |
| Title                                                                                                                                                                   | Applicable Patients(Denominator) | Passed Patients(Numerator) |
| Youth Weight Status Screening (Age >= 3 to < 18)                                                                                                                        | 2                                | 0                          |
| Overweight (Age >= 3 to < 18)                                                                                                                                           | 2                                | 0                          |
| Adult Weight Status Screening (Age >= 18)                                                                                                                               | 5                                | 0                          |
| Adult Weight Screening and Intervention (Age >= 18)                                                                                                                     | 5                                | 0                          |
| Tobacco use screening (Age >= 18)                                                                                                                                       | 5                                | 0                          |
| Tobacco Use Screening and Cessation Intervention (Age >= 18)                                                                                                            | 5                                | 0                          |
| Persistent Asthma treatment on controller meds (Age >= 5 to < 65)                                                                                                       | 0                                | 0                          |
| Persistent Asthma management list need controller meds (Age >= 5 to < 65)                                                                                               | 0                                | 0                          |
| Coronary Vascular Disease on Lipid Therapy (Age >= 18)                                                                                                                  | 0                                | 0                          |
| Coronary Artery Disease need Lipid Therapy (Age >= 18)                                                                                                                  | 0                                | 0                          |
| Ischemic Vascular Disease on Anti-thrombotics (Age >= 18)                                                                                                               | 0                                | 0                          |
| Ischemic Vascular Disease needs Anti-thrombotics (Age >= 18)                                                                                                            | 0                                | 0                          |
| Hypertension Control (Hypertensive Patients with BP of < 140/90)                                                                                                        | 0                                | 0                          |
| Hypertension Out of Control Case Management List                                                                                                                        | 0                                | 0                          |
| Diabetes with A1C < 8.0                                                                                                                                                 | 0                                | 0                          |
| Diabetes with A1C > 9.0                                                                                                                                                 | 0                                | 0                          |
| Cervical Cancer Screening Coverage (Past 2 years) (Age >= 24 to < 65)                                                                                                   | 1                                | 0                          |
| Depression Screening (Age >= 12)                                                                                                                                        | 7                                | 1                          |

## Customized UDS Report in PCHC OpenEMR

**Customized Encounter Forms:** The forms that were used for creating encounter was analyzed and an exact replica of the paper based form was created within the OpenEMR such that the user would have the similar experience as he did with the paper based encounter form. This helped the front office and triage users to have a user friendly screen displaying all data similar to the paper form.

## Customized Patient Encounter form in PCHC OpenEMR

**New Encounter Form**



**Consultation Brief Description:**

**Visit Category:** -- Select One --  
**Facility:** Your Clinic Name Here  
**Billing Facility:** Your Clinic Name Here  
**Sensitivity:** Normal  
**Date of Service:** 2016-12-01   
**Onset/hosp. date:** 

**Issues (Injuries/Medical/Allergy)**   

P: 2016-09-07 Essential (primary) hypertension

*To link this encounter/consult to an existing issue, click the desired issue above to highlight it and then click [Save]. Hold down [Ctrl] button to select multiple issues.*

**Syndromic Surveillance**(Check atleast one box):  
☐ Diarrhoea(3 or more loose BMs in one day)  
☐ Influenza-Like Illness (Sudden onset of fever (>38°C) plus cough and/or sore throat)  
☐ Prolonged Fever (An illness with 3 days or more of high fever)  
☐ Acute Fever & Rash(AFR) (Acute fever with acute non blistering rash)  
☐ No Syndrome (Patient does not have any of syndromes listed above)

**Activity**  
☐ Dietary/physical activity prescription was made for overweight/obesity(HCPC - 0447)  
☐ Referred for Nutritional counseling(HCPC - 0447)  
☐ Tobacco cessation prescription was given to the patient  
☐ Referred for tobacco cessation counselling  
☐ Depression Rx/Counselling performed (GZ58ZZZ)

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## Customized Forms:

Based on the paper format that was used for assisting the patients, PCHC OpenEMR was further customized by adding forms related to Prenatal forms, Past Obstetric forms, Depression Scoring Forms, forms related to Family Planning and Physical exercise forms. These forms were further supported for print options and linked with the reports as required by PCHC.

### Customized form specific to PCHC OpenEMR

| Depression Scoring                                                                                                                                                     |                       |                       |                             |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------------|-----------------------|
| Please read each item below and indicate which language you prefer or understand most                                                                                  | Not at all (0)        | Several days (1)      | More than half the days (2) | Nearly every day (3)  |
| Little interest or pleasure in doing things                                                                                                                            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Feeling down, depressed, or hopeless                                                                                                                                   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Trouble falling or staying asleep, or sleeping too much                                                                                                                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Feeling tired or having little energy                                                                                                                                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Poor appetite or overeating                                                                                                                                            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Feeling bad about yourself- or that you are a failure or have let yourself or your family down                                                                         | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Trouble concentrating on thing, such as reading the newspaper or watching television                                                                                   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Moving or speaking so slowly that other people could have noticed. Or the opposite-being so fidgety or restless that you have been moving around a lot more than usual | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Thoughts that you would be better off dead, or of hurting yourself                                                                                                     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| Add Columns                                                                                                                                                            | 0                     | 3                     | 2                           | 9                     |
| Total Score                                                                                                                                                            | 14                    |                       |                             |                       |
| If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?       |                       |                       |                             |                       |
|                                                                                                                                                                        | Not difficult at all  | Some what difficult   | Very difficult              | Extremely difficult   |
|                                                                                                                                                                        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>       | <input type="radio"/> |
| <div>Save Don't Save</div>                                                                                                                                             |                       |                       |                             |                       |
| Total Score                                                                                                                                                            |                       | Interpretation        |                             |                       |
| 0 - 4                                                                                                                                                                  |                       | Minimal Depression    |                             |                       |
| 5 - 9                                                                                                                                                                  |                       | Mild Depression       |                             |                       |
| 10 - 14                                                                                                                                                                |                       | Moderate Depression   |                             |                       |
| 15 - 27                                                                                                                                                                |                       | Severe Depression     |                             |                       |

## Benefits

The cumbersome process involved in documenting the patients name, maintaining the patients records for years, complexity in searching for the patients name from a group of paper records and accessing the same were reduced by storing all the patient related data in electronic format. The training provided to the doctors and clinicians assisted them to make use of the PCHC OpenEMR in an intuitive and user-friendly manner by providing the exact look and feel similar the paper forms. The generation of reports which earlier, took weeks together in finding the appropriate report, manually counting and processing the patients were reduced with the custom reports that was provided by ViSolve. The PCHC Doctors and Clinicians were happy with the proposed solution and demonstrated increased interested to work with OpenEMR, further exploring the possibilities to use the electronic application more efficiently to improve healthcare outcomes.

## About ViSolve:

- ViSolve is a niche software product development and consulting organization with high value onshore and remote teams with years of experience in Healthcare, Cloud, Web stack and Open Source Technologies. Established in 1995, ViSolve is headquartered in San Jose, California with an offshore development center in India with 50+ Engineers and led by a group of Industry Veterans and Seasoned Professionals.
- With 15+ years of experience in Open Source solutions customization, development and support, ViSolve is one of the leading contributors to the open source community. It has built, released and supported over 60 Open Source products for a leading system vendor in the world.
- ViSolve has strong relationship with HP/ VMWare's product development, QA, Performance, Global pre-sales teams and specializes in Product development, Build, Performance, QA and support of applications and kernel modules on VMware, HP-UX, Linux platforms.
- As the Vice Chair of OEMR Board (a leading clinical solution), ViSolve continues to contribute significantly to OpenEMR code and serve several clinics/hospitals across the globe.
- Has migrated over 160 healthcare practices to Amazon Cloud (AWS), with no business downtime for customer. Further, ViSolve enabled a popular chain of Physical Therapy clinics based in the Bay Area, with 7x24 Business Continuity through High Availability/Disaster Recovery solutions.

To know how ViSolve Healthcare Solutions and Services can help you solve your Business Challenges, please visit our website [hc.visolve.com](http://hc.visolve.com) or email [services@visolve.com](mailto:services@visolve.com) or call +1 (408) 850 2243.

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